Swaption model		1	into	4		
Time	Swap rate	zc	Period	DF(0,t)		
1	1.000%	1.000%	0x1		d_1	1.2920
2	7.000%	7.223%	1x2	0.8698	d_2	1.0920
3	7.000%	7.149%	2x3	0.8129		
4	7.000%	7.112%	3x4	0.7597		
5	7.000%	7.089%	4x5	0.7100		
F	8.884%		A =	3.1525		
σ	20.0%				$XN(-d_2)$ - $FN(-d_1)$	0.000897
T	1		Notional	100,000,000	Receiver swaption	282,632
X	7.000%				(in bps)	28.3

Swaption model		2	into	3		
Time	Swap rate	zc	Period	DF(0,t)		
1	1.000%	1.000%	0x1		d_1	0.1414
2	7.000%	7.223%	1x2		d_2	-0.1414
3	7.000%	7.149%	2x3	0.8129		
4	7.000%	7.112%	3x4	0.7597		
5	7.000%	7.089%	4x5	0.7100		
F	7.000%		A =	2.2826		
σ	20.0%				$XN(-d_2)$ - $FN(-d_1)$	0.007872
T	2		Notional	100,000,000	Receiver swaption	1,796,992
X	7.000%				(in bps)	179.7

Figure S.11: An Example in which a 1 into 4 7% Receiver is Worth Less Than a 2 into 3 7% Receiver

a) As shown in the top half of Figure S.12, the value of a 2 into 3 payer swaption struck at 7% is 324.6 bps, or \$3,246,125 on a notional of \$100mm.

b) As also shown in the top half of Figure S.12, the value of a 2 into 3 receiver swaption struck at 7% is 187.4 bps, or \$1,874,316 on a notional of \$100mm.

c) From swaption parity, we know that the value of the 2 into 3 7% payer swaption less the 2 into 3 7% receiver swaption must be equal to the value of paying 7% in a 2x5 swap. As shown in the bottom half of Figure S.12, the swap is worth \$1,371,809, which is equal to \$3,246,125 − \$1,874,316, the difference between the value of the payer swaption and the receiver swaption.

6. a) In accordance with Equation 5.21, we will need as an input in the pricing of the swaption the underlying forward swap rate. In a calculation not shown here, the 1x4 swap rate is 6.030%. In addition, the annuity factor, A, is 2.5523. The appropriate vol to use is 124.0 bps.

As shown in Figure S.13, for the payer swaption we have

$$d = \frac{F - X}{\sigma_N \sqrt{T}} = \frac{0.0603 - 0.07}{0.0124 \times 1} = -0.7824.$$

From Equation 5.20, the price of the 1 into 3 payer swaption struck at 7% is

$$\begin{aligned}
&A\sigma_N\sqrt{T}\left[\frac{1}{\sqrt{2\pi}}e^{-d^2/2} + dN(d)\right] \\
&\quad = 2.5523 \times 0.0124 \times \sqrt{1} \times \left[\frac{1}{\sqrt{2\pi}}e^{-0.7824^2/2} - 0.7824 \times N(-0.7824)\right] \\
&\quad = 0.00392,
\end{aligned}$$

or 39.2 bps of the notional. On a notional of \$50mm, the premium is \$50mm × 39.2 bps = \$196,199.

Swaption model		2	into	3		
Time	Swap rate	zc	Period	DF(0,t)		
1	6.000%	6.000%	0x1		d_1	0.4050
2	6.250%	6.258%	1x2		d_2	0.0302
3	6.500%	6.522%	2x3	0.8273		
4	6.750%	6.794%	3x4	0.7688	$FN(d_1)$ - $XN(d_2)$	0.014073
5	7.000%	7.075%	4x5	0.7105	Payer swaption	3,246,125
					(in bps)	324.6
F	7.595%		A =	2.3066		
σ	26.5%				$XN(-d_2)$ - $FN(-d_1)$	0.008126
T	2		Notional	100,000,000	Receiver swaption	1,874,316
X	7.000%				(in bps)	187.4

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	6.000%	6.000%	0	0	0	0
2	6.258%	6.516%	0	0	0	0
3	6.522%	7.053%	(7,000,000)	(5,791,333)	7,052,703	5,834,936
4	6.794%	7.614%	(7,000,000)	(5,381,588)	7,613,820	5,853,492
5	7.075%	8.205%	(7,000,000)	(4,973,502)	8,205,211	5,829,804
				(16,146,422)		17,518,232
		Fixed rate	7.000%	NPV	1,371,809	
		Notional	100,000,000			

Figure S.12: Value of 2 into 3 7% Payer Swaption and 2 into 3 7% Receiver Swaption (*top half*) and Value of Paying 7% in a 2×5 Swap (*bottom half*)

Swaption model		1	into	3		
Time	Swap rate	zc	Period	DF(0,*t*)		
1	5.000%	5.000%	0x1		*d*	-0.7824
2	5.250%	5.257%	1x2	0.9026		
3	5.500%	5.519%	2x3	0.8512	$A\sigma_N\sqrt{T}\left[\frac{1}{\sqrt{2\pi}}e^{-d^2/2}+dN(d)\right]$	
4	5.750%	5.787%	3x4	0.7985		
5	6.000%	6.064%	4x5		=	0.00392
F	6.030%		*A* =	2.5523	Payer swaption	196,199
σ_N	0.0124				(in bps)	39.2
T	1		Notional	50,000,000		
X	7.000%					

Figure S.13: Value of 1 into 3 7% Payer Swaption Using the Normal Model

Chapter 6

1. a) The 5-year swap rate is 7.00% as is given in the swap curve.

b) The 5nc2 swap has embedded into it a 2 into 3 year receiver swaption. The swap dealer is the fixed rate payer and is long the receiver swaption. The counterparty is short the option. The cancelable swap rate is 7.646% as shown in Figure S.14. Note that as an input to the calculation of the embedded swaption, we needed the 2x5 swap rate. In a calculation not shown here, the 2x5 swap rate is 7.595%.

c) To determine the appropriate cancelable swap rate, proceed as follows: this position consists of a swap in which the swap dealer pays fixed and receives LIBOR − 25 for five years, and the swap dealer owns an option to receive the same fixed rate and pay LIBOR − 25 for three years, starting in two years. We can model up the swap easily using our swap model, reducing each of the forwards by the spread of 25 bps.

We have to be a little careful computing the value of the swaption. The embedded swaption gives the swap dealer the right to receive a fixed rate, x, and pay LIBOR − 25. Unfortunately, our swaption model only values swaptions

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	6.000%	6.000%	(7,646,209)	(7,213,404)	6,000,000	5,660,377
2	6.258%	6.516%	(7,646,209)	(6,772,114)	6,516,291	5,771,365
3	6.522%	7.053%	(7,646,209)	(6,325,962)	7,052,703	5,834,936
4	6.794%	7.614%	(7,646,209)	(5,878,392)	7,613,820	5,853,492
5	7.075%	8.205%	(7,646,209)	(5,432,633)	8,205,211	5,829,804
				(31,622,506)		28,949,974
	Receiver strike		7.646%		Receiver	2,672,531
	Fixed rate		7.646%			
	Notional		100,000,000	NPV	(0)	

Swaption model		2	into	3		
Time	Swap rate	zc	Period	DF(0, *t*)		
1	6.000%	6.000%	0x1		d_1	0.1694
2	6.250%	6.258%	1x2		d_2	-0.2054
3	6.500%	6.522%	2x3	0.8273		
4	6.750%	6.794%	3x4	0.7688		
5	7.000%	7.075%	4x5	0.7105		
F	7.595%		A =	2.3066		
σ	26.5%				$XN(-d_2) - FN(-d_1)$	0.011586
T	2		Notional	100,000,000	Receiver swaption	2,672,531
X	7.646%				(in bps)	267.3

Figure S.14: Calculating the 5nc2 Swap Rate

in which one counterparty pays LIBOR flat. However, this swaption has the same value as a swaption in which the swap dealer has the right to receive the fixed rate $x + 25$ bps and pay LIBOR flat.[1] Making this adjustment to

[1] If the swaption that has a fixed rate of x versus LIBOR − 25 did not cost the same as the swaption that has a fixed rate of x+25 versus LIBOR flat, we could buy the cheaper of

Time	zc	Forward	Fixed	PV Fixed	Float – 25	PV(Float – 25)
1	6.000%	6.000%	(7,396,209)	(6,977,555)	5,750,000	5,424,528
2	6.258%	6.516%	(7,396,209)	(6,550,693)	6,266,291	5,549,945
3	6.522%	7.053%	(7,396,209)	(6,119,129)	6,802,703	5,628,102
4	6.794%	7.614%	(7,396,209)	(5,686,193)	7,363,820	5,661,292
5	7.075%	8.205%	(7,396,209)	(5,255,008)	7,955,211	5,652,179
				(30,588,578)		27,916,046
	Receiver strike		7.646%		Receiver	2,672,531
	Fixed rate		7.396%			
	Notional		100,000,000	NPV	0	

Swaption model		2	into	3		
Time	Swap rate	zc	Period	DF(0, t)		
1	6.000%	6.000%	0x1		d_1	0.1694
2	6.250%	6.258%	1x2		d_2	-0.2054
3	6.500%	6.522%	2x3	0.8273		
4	6.750%	6.794%	3x4	0.7688		
5	7.000%	7.075%	4x5	0.7105		
F	7.595%		A =	2.3066		
σ	26.5%				$XN(-d_2) - FN(-d_1)$	0.011586
T	2		Notional	100,000,000	Receiver swaption	2,672,531
X	7.646%				(in bps)	267.3

Figure S.15: The 5nc2 Swap Rate When the Floating Side is LIBOR – 25 bps

the swaption strike allows us to iterate to the correct cancelable swap rate. As shown in Figure S.15, we find the cancelable swap rate is 7.396%. Note

the two and sell the more expensive, taking in money today. Cash flows would exactly cancel out at maturity, implying arbitrage. Thus, the two swaptions must be worth the same.

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	6.000%	6.000%	5,815,009	5,485,857	(6,000,000)	(5,660,377)
2	6.258%	6.516%	5,815,009	5,150,252	(6,516,291)	(5,771,365)
3	6.522%	7.053%	5,815,009	4,810,950	(7,052,703)	(5,834,936)
4	6.794%	7.614%	5,815,009	4,470,569	(7,613,820)	(5,853,492)
5	7.075%	8.205%	5,815,009	4,131,565	(8,205,211)	(5,829,804)
				24,049,192		(28,949,974)
	Payer strike		5.815%		Payer	4,900,782
	Fixed rate		5.815%			
	Notional		100,000,000	NPV	0	

Swaption model		2	into	3		
Time	Swap rate	zc	Period	DF(0,*t*)		
1	6.000%	6.000%	0x1		d_1	0.8999
2	6.250%	6.258%	1x2		d_2	0.5251
3	6.500%	6.522%	2x3	0.8273		
4	6.750%	6.794%	3x4	0.7688	$FN(d_1) - XN(d_2)$	0.021246
5	7.000%	7.075%	4x5	0.7105	Payer swaption	4,900,782
					(in bps)	490.1
F	7.595%		A =	2.3066		
σ	26.5%					
T	2		Notional	100,000,000		
X	5.815%					

Figure S.16: The 5np2 Swap Rate

that this is 25 bps below the 5nc2 swap rate that we computed in part b) above. Why is this?

d) In the 5np2 swap, the swap dealer receives fixed and owns the right to terminate the trade. Therefore, the swap dealer is long an embedded 2 into 3 payer swaption. The 5np2 swap rate is 5.815% as shown in Figure S.16.

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	6.000%	6.000%	(7,750,000)	(7,311,321)	6,000,000	5,660,377
2	6.258%	6.516%	(7,750,000)	(6,864,040)	6,516,291	5,771,365
3	6.522%	7.053%	(7,455,596)	(6,168,262)	7,052,703	5,834,936
4	6.794%	7.614%	(7,455,596)	(5,731,850)	7,613,820	5,853,492
5	7.075%	8.205%	(7,455,596)	(5,297,203)	8,205,211	5,829,804
				(31,372,676)		28,949,974
	Receiver strike		7.456%		Receiver	2,422,701
	Fixed rate		7.456%			
	Notional		100,000,000	NPV	0	

Swaption model		2	into	3		
Time	Swap rate	zc	Period	DF(0, t)		
1	6.000%	6.000%	0x1		d_1	0.2367
2	6.250%	6.258%	1x2		d_2	-0.1380
3	6.500%	6.522%	2x3	0.8273		
4	6.750%	6.794%	3x4	0.7688		
5	7.000%	7.075%	4x5	0.7105		
F	7.595%		A =	2.3066		
σ	26.5%				$XN(-d_2)$ - $FN(-d_1)$	0.010503
T	2		Notional	100,000,000	Receiver swaption	2,422,701
X	7.456%				(in bps)	242.3

Figure S.17: A Step-Down Structure

e) In this trade, the swap dealer pays a fixed rate of 7.75% in the first two years, a fixed rate of x in the last three years, and is long a 2 into 3 receiver swaption struck at x. The level of x is found by iterating so that the trade has zero NPV today. We find a rate of 7.456%, as shown in Figure S.17. Notice that the fixed rate is relatively high in the first two years and is lower in the back years. Such a trade is referred to as a *step-down* structure.

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	6.000%	6.000%	(6,750,000)	(6,367,925)	6,000,000	5,660,377
2	6.258%	6.516%	(6,750,000)	(5,978,357)	6,516,291	5,771,365
3	6.522%	7.053%	(6,750,000)	(5,584,499)	7,052,703	5,834,936
4	6.794%	7.614%	0	0	0	0
5	7.075%	8.205%	0	0	0	0
				(17,930,781)		17,266,678
	Receiver strike		6.750%		Receiver	1,501,162
	Fixed rate		6.750%			
	Notional		100,000,000	NPV	837,059	

Swaption model		1	into	2		
Time	Swap rate	zc	Period	DF(0, t)		
1	6.000%	6.000%	0x1		d_1	0.1768
2	6.250%	6.258%	1x2	0.8857	d_2	-0.1542
3	6.500%	6.522%	2x3	0.8273		
4	6.750%	6.794%	3x4			
5	7.000%	7.075%	4x5			
F	6.775%		A =	1.7130		
σ	33.1%				$XN(-d_2) - FN(-d_1)$	0.008763
T	1		Notional	100,000,000	Receiver swaption	1,501,162
X	6.750%				(in bps)	150.1

Figure S.18: Valuing the 3nc1 Single European Struck at 6.75%

2. If we do not cancel the trade today, we will be left paying fixed for three years at a rate of 6.75% with the right to cancel in one year. That is, we will be paying fixed in a 3nc1 single European at a rate of 6.75%. As shown in Figure S.18, the NPV of this trade is $837,059. Since this is a positive amount, we should not cancel the trade. Note that in computing the value of the 3nc1, we have made use of 1 into 2 swaption vol of 33.1% and also made use of the 1x3 swap rate of 6.775% (calculation not shown here).

4. a) A 1 into 3 receiver swaption is embedded into the trade. The 1 into 3 lognormal implied vol for the at-the-money forward strike is 30.6%.

b) The level of normal implied vol for the at-the-money forward strike is given by

$$\begin{aligned}\sigma_N &\approx F \times \sigma_{LN} \times \left(1 - \frac{\sigma_{LN}^2 T}{24}\right)\\ &= 0.07035 \times 0.306 \times \left(1 - \frac{0.306^2 \times 1}{24}\right)\\ &= 0.02144,\end{aligned}$$

or 214.4 bps.

c) As shown in Figure S.19, the 4nc1 cancelable swap rate is 7.638% according to the lognormal model.

d) As shown in Figure S.20, the 4nc1 cancelable swap rate is 7.593% according to the normal model.

e) The 4nc1 cancelable swap rate is higher using the lognormal model than it is using the normal when we (inconsistently) assume flat skew for both. Of course, the assumption of flat skew in the normal model implies downward lognormal skew. Under the assumption of flat normal skew, if we had instead used the lower level of lognormal implied vol for this high-strike option that the normal model necessitates, we would come out with the same lower cancelable swap rate of 7.593% that we solved for in part d).

7. a) As shown in Figure S.21, the extendible swap rate is 7.638%.

b) This is the same rate we solved for in problem 4 c) for a 4nc1. This is no coincidence; the two trades are identical, a direct result of swaption parity, as we will now show.

Recall that by swaption parity we know that being long a 1 into 3 7.638% receiver and short a 1 into 3 7.638% payer is equivalent to receiving fixed at 7.638% in a 3-year swap, one year forward. This implies

$$\begin{aligned}&\text{Long 1 into 3 payer struck at 7.638\%}\\ &\quad = \text{Long 1 into 3 receiver struck at 7.638\%} +\\ &\qquad \text{Paying fixed at 7.638\% in a 1x4.}\end{aligned} \tag{S.1}$$

We will use Equation S.1 to substitute into one of the lines below.